Entropy games in quasipolynomial time, conditional on a labelling theorem for discounted monotone maps

Worasait Suwannik*

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Abstract

Entropy games, introduced by Asarin, Cervelle, Degorre, Dima, Horn and Kozyakin, are played by Despot and Tribune against a nondeterministic People; the value is a growth rate, an algebraic number. Comparing the value with a rational number is in NP $\cap$ coNP, and polynomial-time algorithms are known only under restrictions: when Despot has a bounded number of states with a choice, or, for unweighted games, when a rank parameter is bounded (for weighted games of bounded rank the known bound is pseudo-polynomial). We show that the threshold problem "is the value at state d at least r?", for positive rational weights and a rational r written in binary, is solvable deterministically in time $2^{O((\log(L+2))^2)}$, where L is the input length. The result is conditional: the algorithm uses as a black box a recursive labelling procedure for discounted monotone maps, and we assume the labelling theorem for this procedure (a theorem on its correctness and two lemmas on its branching) that is stated in a recent unrefereed manuscript made public by OpenAI. Since the dynamic programming operator of an entropy game is transcendental, applying the procedure requires three ingredients: (i) an exactly computable surrogate of the log-sum-exp operator that is exactly monotone and exactly commutes with translations; (ii) a quantitative bound on the distance between the discounted fixed point and the value, valid for reducible matrices; (iii) a separation bound at a rational threshold. These three ingredients do not depend on the assumed theorem; they are proved in full, using from the literature only the existence of the value and of optimal positional strategies. Polynomial time remains open: we show that, on every instance of our algorithm, the recursion on which it relies makes a number of calls that is superpolynomial in the number of Despot states.

1 Introduction

Entropy games. An entropy game [1, 2] is played on a finite graph whose nodes are split into states of Despot, of Tribune, and of People. Despot and Tribune choose moves; People is nondeterministic. A pair of strategies determines a set of possible histories; Tribune wants the (weighted) number of histories to grow as fast as possible with the horizon, Despot wants to slow it down. The value at an initial state is the exponential growth rate. Asarin et al. proved that entropy games are determined with simple optimal strategies and that their solution is in NP $\cap$ coNP [1]. Akian, Gaubert, Grand-Clément and Guillaud [2] gave an operator approach, proved that the value exists for every initial state with optimal positional strategies, and showed that entropy games in which Despot has a fixed number of significant states, that is, states in which he has at least two moves, can be solved in polynomial time. Allamigeon, Gaubert, Katz and Skomra [3] obtained polynomial and pseudo-polynomial bounds when the rank of the game is fixed, the rank being the maximum of the ranks of the weighted transition matrices

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determined by the pairs of strategies of the two players (their "ambiguity matrices"), and state in their introduction: "The complexity of entropy games without restrictions on the number of (significant) Despot positions is an open problem." The same question is recalled in [4]: "it is an open question whether general entropy games can be solved in polynomial time"; that paper gives parameterised bounds for irreducible entropy games.

The result. Let D be the set of Despot states, and let L be the length of a binary encoding of the game and of the threshold (Section 2).

Theorem 1.1. *Assume Assumption A. There are a deterministic multitape Turing machine and an absolute constant C such that, for every entropy game with positive rational weights and every positive rational r, all given in binary, the machine outputs the set*

$$\{\, d \in D : \; V_d^{\infty} \geq r \,\}$$

within $2^{C(\log_2(L+2))^2}$ steps, where V_d^{∞} is the value of the game with initial state d.

By [2, Proposition 11] the value of the original game of Asarin et al., in which the initial state is free, is $\max_d V_d^{\infty}$, so the same bound holds for comparing that value with r. The value V_d^{∞} lies between the least weight and the sum of all weights (Section 2), hence between 2^{-L} and 2^{L}. A binary search over thresholds r of $O(L+k)$ bits, first over the powers of two and then by bisection of an interval $[2^j, 2^{j+1}]$, therefore gives V_d^{∞} to within any prescribed relative precision 2^{-k} in time quasipolynomial in $L+k$. Optimal strategies are not computed (Remark 8.1).

The assumption. Theorem 1.1 assumes Assumption A (Section 3), which collects three statements of the manuscript [9], a theorem and two lemmas: they assert the termination, the correctness and the branching bounds of a recursive labelling procedure, which our algorithm calls as a black box. We refer to the three statements together as the *labelling theorem*. The manuscript, dated October 5, 2026, belongs to the OpenAI Math release [11], a public repository of manuscripts attributed to OpenAI. It is, to our knowledge, unrefereed, and we have not re-verified the proof of the statements we take from it; the formal proofs listed in [11] do not include these statements. Everything else in the paper is proved in full, using in addition one theorem from the entropy-games literature (existence of the value and of optimal positional strategies, [2, Theorem 10]); in particular Sections 5, 6 and 7 do not depend on [9]. The correctness part of Theorem 1.1, with the results it rests on, has in addition been formalised in Lean, with Assumption A and the theorem of [2] as hypotheses; the running-time part has not (Appendix C).

Relation to mean-payoff games. Structurally, entropy games relate to deterministic mean-payoff games as stochastic mean-payoff games do, with a multiplicative aggregation over People's moves in place of an expectation; in the one-player case this is the multiplicative criterion of risk-sensitive Markov decision processes and of multiplicative Markov decision chains (see the references given in [2, proof of Proposition 1]). Deterministic entropy games, in which People has a single action in every state, can be re-encoded as deterministic mean-payoff games [2, Remark 3]; the weights of the resulting mean-payoff game are the logarithms of the weights of the entropy game. In the other direction, Asarin et al. transform a mean-payoff game with nonnegative integer weights w into a deterministic weighted entropy game with weights 2^w and remark that, "due to the exponential encoding of payoffs", the complexity obtained in this way is not as good as that of direct algorithms [1, Section 6.2]. For weights written in binary the transformation is exponential. *Consequently Theorem 1.1 contains mean-payoff games only for weights written in unary*, for which polynomial algorithms were already known (as recalled in [1, Section 6.2]); it does not re-prove, and is not a generalisation of, the theorem of [10] on mean-payoff games with binary weights.

The labelling theorem of [9]. The manuscript [9] states that turn-based stochastic mean-payoff games with binary data are solvable in deterministic quasipolynomial time. Its engine is an abstract statement about a map $F : \mathbb{R}^n \to \mathbb{R}^n$ that is monotone and satisfies $F(t + c\mathbf{1}) = F(t) + \gamma c\mathbf{1}$ with $0 < \gamma < 1$: a recursive procedure, which evaluates only $\lfloor F(t) \rfloor$ at integer vectors t, labels the coordinates of an integer box with signs in such a way that every integer subsolution near the top of the box and every integer supersolution near its bottom is respected. In [9] it is applied to a scaled discounted operator of the stochastic game, which is piecewise affine with rational coefficients, so that $\lfloor F(t) \rfloor$ is computed exactly.

Contributions. For entropy games the dynamic programming operator is, in logarithmic coordinates,

$$f_d(t) = \min_{(d,\vartheta)\in E} \max_{(\vartheta,p)\in E} \log \sum_{(p,d')\in E} m_{pd'}\, e^{t_{d'}},$$

where E is the set of arcs of the game and the $m_{pd'}$ are its weights (Section 2), and three things stand in the way. They are the contributions of this paper.

(i) *Exact arithmetic for a transcendental operator* (Section 5). The labelling theorem needs monotonicity and the translation identity exactly; an approximation of $\log \sum \exp$ computed with rounding has neither. We replace exp by a monotone rational function E and define the surrogate of $\log \sum_j m_j e^{t_j}$ as the unique root s of $\sum_j m_j E(t_j - s) = 1$. Because the test "$\sum_j m_j E(t_j - s) \geq 1$" is upward closed in t and depends on $t - s\mathbf{1}$ only, the surrogate is exactly monotone and exactly translation-equivariant *for any continuous, strictly increasing E from $\mathbb{R}$ onto $(0, \infty)$* (Lemma 5.2); accuracy is a separate, easy matter. The floor of the surrogate is found by exact comparisons of rationals of polynomial size.

(ii) *A quantitative discount bound* (Section 6). For a nonnegative matrix M with nonzero rows and $\gamma = 1 - 1/N$, the fixed point v of $t \mapsto \gamma \log(Me^t)$ satisfies $|v_i/N - \lambda_i| \leq R/N$, where λ_i is the logarithm of the largest Perron root of a class accessible from i and $R = O(n \log N + n^2\Omega)$, with Ω the logarithmic size of the entries. The bound holds for reducible matrices, including chains of classes with equal Perron roots; the price of reducibility is the additive $\log N$ per level, not a spectral gap.

(iii) *A separation bound at a rational threshold* (Section 7). After normalisation to the threshold 1, the logarithm of the value is either 0 or at least $1/H$ in absolute value, where H is an integer with polynomially many bits; the proof is elementary.

Section 4 derives from Assumption A the fixed-point localisation procedure in the generality we need. The manuscript [9] carries out the same box-shrinking argument for its own map; we give the argument in full for a general map. Section 8 assembles the algorithm (Algorithm 1); its structure is shown in Figure 1.

Relation to known results. Several ingredients are known or routine. Proposition 4.3 and Lemmas 4.4, 4.5 restate, for a general map, [9, Section 4]. Lemma 6.1 is textbook material on nonnegative matrices, proved here to keep the paper self-contained, and Lemma 7.1(b) is the standard min–max representation of a discounted fixed point. The separation bound, Lemma 7.2, is a routine variant, at a rational threshold, of the separation bounds between two values in [2, Corollary 29] and [3]. To our knowledge the surrogate (Lemmas 5.2 and 5.3, Proposition 5.5) and the discount bound (Theorem 6.2, with Remark 6.3) are new, and so is Theorem 1.1; Proposition 9.1 is a simple observation, also new to our knowledge.

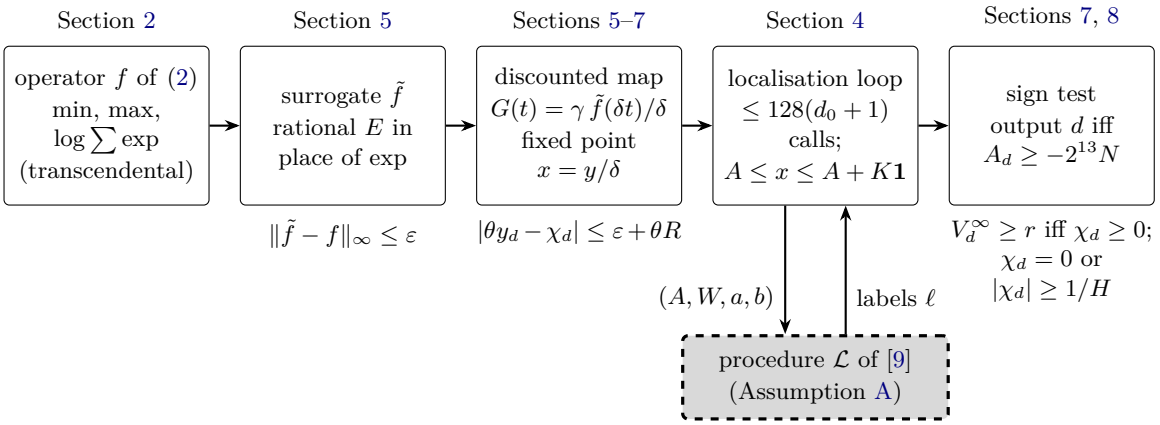


Figure 1: Structure of the algorithm (Algorithm 1) and of the proof of Theorem 1.1. Here $\chi_d = \ln V_d^\infty$ after normalisation to the threshold 1, $\theta = 1/N$, $\gamma = 1-\theta$, and y is the fixed point of $\gamma\tilde f$; the parameters δ, K, H, ε, N, d_0 are those of Table 1, and R is the constant of Theorem 6.2. Each of the three errors ε, θR and $\theta\delta K$ is at most $1/(64H)$, so that $\theta\delta A_d$ is within $1/(16H)$ of χ_d, whereas a nonzero χ_d is at least $1/H$ in absolute value. The shaded, dashed box is the only component whose properties are taken from [9] without proof (Assumption A); the guarantee of the localisation loop depends on it.

Comparison with the direct approaches. Three routes suggest themselves, and none of them works without the ingredients above. (a) The known reduction of entropy games is to Kullback–Leibler games, which are stochastic mean-payoff games in which some action spaces are simplices and the payments are relative entropies [2]. These have infinite action spaces, whereas [9, Theorem 1.1] is for finite turn-based games; we know of no discretisation of the simplices of polynomial size that preserves the answer at the precision $2^{-\mathrm{poly}(L)}$ required by the separation bound. (b) The re-encoding of deterministic entropy games as mean-payoff games produces irrational weights, the logarithms of the weights of the entropy game, and applies only when People has no choice. (c) Applying the labelling theorem directly to the discounted operator γf requires the exact value of $\lfloor \gamma f(t) \rfloor$ at integer points, that is, the sign of $\sum_j m_j e^{x_j} - 1$ at rational x_j; we know of no polynomial-time procedure for this. For a related objective (entropic risk with total reward in turn-based stochastic games), where exponentials of the data also enter the equations to be decided, the threshold problem is, in the most general case, only known to be decidable subject to Schanuel's conjecture [5]. Moreover the quantitative discount estimate [9, Theorem 2.1] concerns expectations and does not give the vanishing-discount estimate for the multiplicative criterion that Theorem 6.2 provides.

Limits and open problems. The result is a complexity bound; we make no claim of practical efficiency. Polynomial time for entropy games is open, as it is for mean-payoff and simple stochastic games. Section 9 records a simple observation: the recursion of [9], as written, makes at least $\binom{d-11+B_0}{B_0}$ calls on every input, where 2^d is the order of the box width and $B_0 = \Theta(\log n)$; for our algorithm this is superpolynomial in the number n of Despot states on every instance (Proposition 9.1); it says nothing when n is bounded. A polynomial bound would therefore require a change in the recursion itself. We also leave open the computation of optimal strategies within the same bound, and the extension to operators mixing expectations and log-sum-exp gates (risk-sensitive stochastic games; compare [5] for the total-reward objective), for which the surrogate of Section 5 applies verbatim but the analogue of Section 6 is missing, and the analogue of the separation bound of Section 7 meets an obstacle from number theory (Section 9).

Other related work. The quasipolynomial lower bounds for separating automata and universal trees of Czerwiński, Daviaud, Fijalkow, Jurdziński, Lazić and Parys [6], and the result of Fijalkow, Gawrychowski and Ohlmann that separating automata do not yield a quasipolynomial algorithm for mean-payoff games [8], concern a different class of algorithms; the procedure of [9] is not of that form. The manuscript [10], whose recursion is the one adapted in [9], makes a remark of this kind about its own recursion, with respect to the lower bounds for universal graphs of [8] only. The relation between the procedure and these lower bounds is not studied here.

2 Entropy games, encoding, and the threshold problem

We use the model of [2, Section 2]. The set of nodes is a disjoint union $D \cup T \cup P$ (states of Despot, Tribune, People), the set of arcs E is contained in $(D \times T) \cup (T \times P) \cup (P \times D)$, every node has at least one outgoing arc, and a weight $m_{pd} > 0$ is attached to every arc $(p, d) \in P \times D$; the other arcs have weight one. A turn consists of a move of Despot, a move of Tribune and a move of People. A *history* is a finite path of the graph starting at the initial state. As in [2, Section 2], a *strategy* of Despot is a map that assigns to every history ending in a state $d \in D$ an arc $(d, \vartheta) \in E$, and a strategy of Tribune is a map that assigns to every history ending in a state $\vartheta \in T$ an arc $(\vartheta, p) \in E$; thus strategies are deterministic and may depend on the whole history. For strategies σ of Despot and τ of Tribune and an initial state d, the payment $R_d^k(\sigma, \tau)$ in horizon k is the sum of the weights (products of arc weights) of all paths of k turns starting at d and compatible with σ, τ, and $R_d^\infty(\sigma, \tau) = \limsup_k R_d^k(\sigma, \tau)^{1/k}$. The sequence $R_d^k(\sigma, \tau)^{1/k}$ is bounded (it lies between the least weight and the sum of all weights), so the lim sup is finite. Tribune maximises, Despot minimises; the value V_d^∞ and optimal strategies are defined by the saddle-point condition $R_d^\infty(\sigma, \tau^*) \geq V_d^\infty \geq R_d^\infty(\sigma^*, \tau)$ for all strategies σ, τ. A *policy* is a stationary positional strategy: a map $\sigma : D \to T$ with $(d, \sigma(d)) \in E$, respectively $\tau : T \to P$ with $(\vartheta, \tau(\vartheta)) \in E$. The dynamic programming operator is

$$\mathcal{F}_d(X) = \min_{(d,\vartheta)\in E} \max_{(\vartheta,p)\in E} \sum_{(p,d')\in E} m_{pd'} X_{d'} \qquad (X \in \mathbb{R}^D_{>0},\ d \in D). \tag{1}$$

Theorem ([2, Theorem 10])**.** "The infinite horizon entropy game has a value and it has optimal positional strategies, namely the Blackwell optimal strategies $(\delta^*, \tau^*) \in \mathcal{P}_D \times \mathcal{P}_T$ of Theorem 8. Moreover, for all initial states d, $V_d^\infty = \lim_{k\to\infty}(V_d^k)^{1/k}$."

Here $\mathcal{P}_D, \mathcal{P}_T$ are the sets of policies, δ^* is the policy of Despot that we denote by σ^* (the letter δ has another use from Section 5 on), "Theorem 8" is [2, Theorem 8], $V^k = \mathcal{F}^k(\mathbf{1})$ by [2, Proposition 1], where $\mathbf{1}$ is the all-ones vector (written e in [2]), and the weights are arbitrary positive reals. Optimality is meant against all strategies of the opponent, not only policies: in [2, Section 2] strategies are maps defined on histories and the saddle-point condition, Equation (3) there, is required "for all strategies δ, τ of Despot and Tribune"; accordingly, in the proof of [2, Theorem 10] the inequality following Equation (22) is stated "for all strategies τ of Tribune, non necessarily positional", and the one following Equation (23) "for all strategies δ of Despot, non necessarily positional".

Encoding and model of computation. The sets D, T, P are initial segments of the positive integers; the input lists the arcs, each weight m_{pd} as a pair of positive integers (numerator, denominator) in binary, and the threshold $r > 0$ in the same way. L is the total number of bits. Thus $n := |D| \leq L$ and the number of arcs is at most L. Running time is the number of steps of a deterministic multitape Turing machine. The *threshold problem* is to output $\{d \in D : V_d^\infty \geq r\}$.

Normalisation. Multiplying all weights by a constant $c > 0$ multiplies every R_d^k by c^k and hence every R_d^∞ and V_d^∞ by c. Write $m_e = b_e/c_e$ for the arcs $e \in E \cap (P \times D)$ and $r = r_1/r_2$, and put

$$q = r_1 \prod_e c_e, \qquad a_e = b_e\, r_2 \prod_{e' \neq e} c_{e'}, \qquad \text{so that} \quad \frac{m_e}{r} = \frac{a_e}{q}.$$

Then q and $a_{\max} := \max_e a_e$ are positive integers with at most L bits. (One may take the least common multiple of r_1 and the c_e instead of their product; no bound below changes.) *From now on the weights are $m_e = a_e/q$ and the threshold is* 1. We write

$$\chi_d = \ln V_d^\infty, \qquad \Omega = \ln\bigl(n \max(a_{\max}, q)\bigr),$$

and we have to decide, for every d, whether $\chi_d \geq 0$. Note $1/q \leq m_e \leq a_{\max}$.

Logarithmic coordinates. For $t \in \mathbb{R}^D$ and a People state p let $\ell_p(t) = \ln \sum_{(p,d') \in E} m_{pd'} e^{t_{d'}}$, and

$$f_d(t) = \min_{(d,\vartheta) \in E} \max_{(\vartheta,p) \in E} \ell_p(t), \qquad f = \log \circ \mathcal{F} \circ \exp\,. \tag{2}$$

For policies σ, τ let $M^{\sigma\tau} \in \mathbb{R}_{\geq 0}^{D \times D}$ be the matrix with $M^{\sigma\tau}_{dd'} = m_{pd'}$ if $(p, d') \in E$, where $p = \tau(\sigma(d))$, and 0 otherwise. Every row of $M^{\sigma\tau}$ is nonzero, its positive entries lie in $[1/q, a_{\max}]$, and $R_d^k(\sigma, \tau) = \bigl((M^{\sigma\tau})^k \mathbf{1}\bigr)_d$.

3 The labelling theorem and the assumption

This section quotes the statements of [9, Section 3] ("Simultaneous labeling for discounted monotone maps") that we use, verbatim (two displayed formulas of Definition 3.1 are set inline) and with the numbering of [9]; omissions are marked by [...]. The quotations keep the notation of [9]. In particular $D = h2^d$ is there the width of a box; in our own text we write W for this width, because D is the set of Despot states, and the letters d, h, P, p, q, M, r, ρ, k, c, u have the meaning of [9] only where the procedure and its recursion tree are discussed (Sections 4 and 9, the running-time analysis of Section 8, and Appendix A).

Setting ([9, Equation (3.1)])**.** "We therefore work in the following abstract setting: $n \geq 1$ and $F : \mathbb{R}^n \to \mathbb{R}^n$ satisfy

$$t \leq t' \implies F(t) \leq F(t'), \qquad F(t + c\mathbf{1}) = F(t) + \gamma c \mathbf{1} \quad (t \in \mathbb{R}^n,\ c \in \mathbb{R}), \qquad 0 < \gamma < 1.$$

All vector inequalities, minima, maxima, and floors in this section are coordinatewise. The algorithm uses F only by evaluating $\lfloor F(t) \rfloor$ at integer vectors t."

Definition (inputs and witnesses; [9, Equation (3.2), Definition 3.1])**.** "An input consists of $A \in \mathbb{Z}^n$, a width

$$D = h2^d, \qquad 128 \leq h \leq 255, \quad h, d \in \mathbb{Z}, \quad d \geq 0,$$

and positive rational mass vectors $a, b \in \mathbb{Q}^n_{>0}$. Write $u = A + D\mathbf{1}$, $P = D/32$, and $a(S) = \sum_{i \in S} a_i$, $b(S) = \sum_{i \in S} b_i$ for $S \subseteq \{1, \ldots, n\}$. [...]"

"An integer vector $y \in [A, u]$ is a *subsolution in the box* if $F_i(y) \geq y_i$ for every $i \in S_y$, $S_y = \{i : y_i > A_i\}$. It is a *plus witness* if also $a(S_y) \leq 1$; its claims are the indices i for which $y_i \geq u_i - P$. An integer vector $z \in [A, u]$ is a *supersolution in the box* if $F_i(z) \leq z_i$ for every $i \in S_z$, $S_z = \{i : z_i < u_i\}$. It is a *minus witness* if also $b(S_z) \leq 1$; its claims are the indices i for which $z_i \leq A_i + P$. [...]"

The omitted sentences are a remark on the uniqueness of the representation $D = h2^d$ and on its later use, the definitions of the terms *support* (for S_y, S_z) and *relevant witness* and, in the last quotation below, the remark that B is used only in the proof and is not computed by the procedure.

Theorem ([9, Theorem 3.2 (Recursive simultaneous labeling)])**.** "Let $F : \mathbb{R}^n \to \mathbb{R}^n$ satisfy Equation (3.1). There is a terminating deterministic procedure $\mathcal{L}(A, D, a, b)$, using exact evaluations of $\lfloor F(t) \rfloor$ for $t \in \mathbb{Z}^n$ and integer and rational arithmetic, with the following guarantee. For every input as in Equation (3.2) and Definition 3.1, its output $\ell \in \{+, -\}^n$ satisfies

$$\begin{aligned} y_i \geq A_i + D - P &\implies \ell_i = + \quad \text{for every plus witness } y, \\ z_i \leq A_i + P &\implies \ell_i = - \quad \text{for every minus witness } z." \end{aligned}$$

The two lemmas quoted next use the vocabulary of the procedure itself, which is reproduced in Appendix A. The procedure is recursive. It has two base cases, the mass base case and the *width base case* (Steps 1 and 2 there). A call to which neither applies, a *nonbase call*, makes recursive calls, its *children*: the calls of its two *pivot passes* (Steps 3 and 4), the *provisional call* (Step 5) and the continuation (Step 6). A *child input* is the input of such a call.

Lemma ([9, Lemma 3.4 (Valid boxes and bounded passes)])**.** "Every child input is valid and its box lies inside its parent's box. If the child calls terminate, each pivot pass makes at most $1024n + 1$ calls. The width base case makes at most $522240n + 1$ evaluations of $\lfloor F \rfloor$."

Lemma ([9, Equation (3.10), Lemma 3.5 (Rank and branching)])**.** "Define $\mu(a, b) = \min_i a_i b_i$, $\rho = \frac{32}{25}$, $B(\mu) = \min\{j \in \mathbb{Z}_{\geq 0} : \rho^j \mu > 1\}$. [...]" "The procedure terminates. At a nonbase call every child has smaller rank $d + B(\mu(a, b))$. There is exactly one child preserving B, the provisional call, and it decreases d by one. There are at most $2048n + 3$ other children; each decreases B by at least one and does not increase d."

Theorem 3.2 of [9] asserts that "there is" a procedure, while its Lemmas 3.4 and 3.5 speak of "the procedure". In [9] the two are identified by the proof of its Theorem 3.2, which shows that the procedure written out in its Section 3.2 has the stated guarantee; this identification is not contained in the three statements themselves, and we make it part of the assumption.

Assumption A (labelling procedure)**.** Let $\mathcal{L}$ denote the procedure of [9, Section 3.2], reproduced in Appendix A. For every map F as in Equation (3.1) of [9]:

(a) $\mathcal{L}$ terminates on every input as in Equation (3.2) and Definition 3.1 of [9], and has the guarantee stated in Theorem 3.2 of [9];

(b) Lemma 3.4 of [9] holds for $\mathcal{L}$;

(c) Lemma 3.5 of [9] holds for $\mathcal{L}$.

Besides Assumption A we use the text of the procedure, to count the operations performed at a node of the recursion (Lemmas 4.4 and 4.5) and for the observation of Section 9. A *call* is a node of the recursion tree of $\mathcal{L}$; the children of a nonbase call are the calls made in its two pivot passes, the provisional call and the continuation.

4 Fixed-point localisation

Throughout this section $N \geq 2$ is an integer, $\gamma = 1 - 1/N$, and $G : \mathbb{R}^n \to \mathbb{R}^n$ satisfies

$$t \leq t' \Rightarrow G(t) \leq G(t'), \qquad G(t + c\mathbf{1}) = G(t) + \gamma c\mathbf{1} \quad (t \in \mathbb{R}^n,\ c \in \mathbb{R}). \tag{3}$$

Lemma 4.1. *A map G satisfying (3) is a γ-contraction for the sup norm; it has a unique fixed point x. If $w \leq G(w)$ then $w \leq x$, and if $w \geq G(w)$ then $w \geq x$.*

Proof. For t, t' put $c = \|t - t'\|_\infty$. Then $t \leq t' + c\mathbf{1}$, so $G(t) \leq G(t') + \gamma c\mathbf{1}$, and symmetrically. Banach's theorem gives x and $G^k(w) \to x$ for every w. If $w \leq G(w)$, monotonicity gives $w \leq G(w) \leq G^2(w) \leq \cdots \to x$. The other case is symmetric. □

The same proof shows: a monotone map Ψ with $\Psi(t + c\mathbf{1}) \le \Psi(t) + \gamma c\mathbf{1}$ for $c \ge 0$ is a γ-contraction, and sub- and supersolutions lie below and above its fixed point. We use this remark in Section 6.

Lemma 4.2 (comparison vectors)**.** *Let x be the fixed point of G and $Y = \lfloor x \rfloor - N\mathbf{1}$, $Z = \lceil x \rceil + N\mathbf{1}$. Then $Y, Z \in \mathbb{Z}^n$, $G(Y) \ge Y$, $G(Z) \le Z$, $Y \le x \le Z$ and $Z_i - Y_i \le 2(N+1)$ for all i.*

Proof. $x - (N+1)\mathbf{1} \le Y \le x - N\mathbf{1}$, hence by (3) $G(Y) \ge G(x) - \gamma(N+1)\mathbf{1} = x - (N - \frac{1}{N})\mathbf{1} \ge x - N\mathbf{1} \ge Y$. Similarly $G(Z) \le x + \gamma(N+1)\mathbf{1} \le x + N\mathbf{1} \le Z$. The rest is clear. □

The *permitted widths* are the integers $W = h2^d$ with $128 \le h \le 255$, $d \ge 0$; for such a $W > 255$ its *predecessor* is $W^- = (h-1)2^d$ if $h > 128$ and $W^- = 255 \cdot 2^{d-1}$ if $h = 128$. In both cases W^- is a permitted width and $0 < W - W^- \le W/128$.

Proposition 4.3 (localisation)**.** *Assume Assumption A(a). Let G satisfy (3) with fixed point x, let J be an integer with $\|x\|_\infty + N + 1 \le J$, and let $K = 256(N+1)$. Consider the following loop. Start with $A = -J\mathbf{1}$ and $W = 128 \cdot 2^{d_0}$, where d_0 is the least nonnegative integer with $128 \cdot 2^{d_0} \ge 2J$. While $W > K$: compute $\ell = \mathcal{L}(A, W, a, b)$ with $a_i = b_i = 1/n$ for the map $F = G$; replace A_i by $A_i + (W - W^-)$ for every i with $\ell_i = +$, leave the other A_i unchanged, and replace W by its predecessor W^-. Then the loop makes at most $128(d_0+1)$ calls of $\mathcal{L}$, and at its end*

$$A \in \mathbb{Z}^n, \qquad A_i \le x_i \le A_i + K \quad \text{for all } i.$$

Proof. Let Y, Z be as in Lemma 4.2. We show that $A \le Y$ and $Z \le A + W\mathbf{1}$ hold throughout. Initially $\|Y\|_\infty, \|Z\|_\infty \le \|x\|_\infty + N + 1 \le J$ and $-J + W \ge J$.

Suppose the invariant holds before an iteration, and put $u = A + W\mathbf{1}$, $P = W/32$. Since $W > K > 255$, the input (A, W, a, b) is of the form required by Equation (3.2) of [9]. The vector Y is an integer vector of $[A, u]$ with $G_i(Y) \ge Y_i$ for all i, so it is a subsolution in the box; its support has a-mass at most $n \cdot \frac{1}{n} = 1$, so it is a plus witness. In the same way Z is a minus witness. By Assumption A(a), $Y_i \ge u_i - P$ implies $\ell_i = +$ and $Z_i \le A_i + P$ implies $\ell_i = -$. Taking contrapositives,

$$\ell_i = + \implies Z_i > A_i + P, \qquad \ell_i = - \implies Y_i < u_i - P.$$

By Lemma 4.2 and $W > K$ we have $Z_i - Y_i \le K/128 < W/128$. Let $\omega = W - W^- \le W/128$. If $\ell_i = +$ then $Y_i > Z_i - W/128 > A_i + W/32 - W/128 = A_i + 3W/128 \ge A_i + \omega$, so the new lower end $A_i + \omega$ is still $\le Y_i$, and the new upper end $A_i + \omega + W^- = u_i$ is unchanged. If $\ell_i = -$ then $Z_i < Y_i + W/128 < u_i - W/32 + W/128 \le u_i - \omega = A_i + W^-$, so the new upper end is still $\ge Z_i$, and the lower end is unchanged. The invariant is preserved.

The widths form a strictly decreasing sequence of permitted widths with exponent at most d_0; there are 128 mantissas per exponent, so there are at most $128(d_0 + 1)$ iterations. At the end $W \le K$ and $A \le Y \le x \le Z \le A + W\mathbf{1} \le A + K\mathbf{1}$. □

The number $P = W/32$ in this proof need not be an integer: for $N = 2$ the permitted width $255 \cdot 2^2$ exceeds $K = 768$. Equation (3.2) and Definition 3.1 of [9], quoted in Section 3, admit every exponent $d \ge 0$ and do not require P to be an integer. In the algorithm of Section 8 the question does not arise: there $N \ge 2^{15}$, so that every width $W = h2^d > K$ has $d \ge 16$, and P is an integer.

Lemma 4.4 (size of one call of $\mathcal{L}$)**.** *Assume Assumption A(b),(c), and refer to the text of the procedure (Appendix A). A call $\mathcal{L}(A, h2^d, a, b)$ with $a_i = b_i = 1/n$ has a recursion tree with at most*

$$(\Lambda{+}1)(B_0{+}1)\,(\Lambda s_*)^{B_0} \quad \text{nodes}, \qquad \Lambda = d{+}B_0, \quad s_* = 2048n{+}3, \quad B_0 = B(1/n^2) \le 2{+}6\log_2 n,$$

all of whose boxes are contained in $[A, A + h2^d\mathbf{1}]$. *Only width base cases evaluate* $\lfloor F \rfloor$*; each of them makes at most* $522240n + 1$ *evaluations of the vector* $\lfloor F(t) \rfloor$*, all at integer points* t *of its own box, hence of the root box.*

Proof. By Lemma 3.5 of [9] the rank $d + B$ strictly decreases along every edge of the tree and is nonnegative, so the depth is at most Λ; and on a root-to-node path at most B_0 edges decrease B, since $B \geq 0$ never increases. A node at depth k whose path has j B-decreasing edges is determined by the positions of these edges and, at each of them, a choice among at most s_* children; the other edges go to the unique B-preserving child. Hence the number of nodes is at most $\sum_{k=0}^{\Lambda} \sum_{j=0}^{\min(k,B_0)} \binom{k}{j} s_*^j \leq (\Lambda+1)(B_0+1)(\Lambda s_*)^{B_0}$ (here $\Lambda \geq B_0 \geq 1$, because $\mu = 1/n^2 \leq 1$). It remains to bound B_0. For $n = 1$, $\mu = 1$ and $B_0 = 1$. For $n \geq 2$, $\rho^3 > 2$ gives $\rho^{6\log_2 n} > n^2$, so $B_0 \leq \lceil 6 \log_2 n \rceil$. In both cases $B_0 \leq 2 + 6 \log_2 n$. Boxes of children lie in the box of the parent (Lemma 3.4 of [9]). In the text of the procedure the map F occurs only in Step 2, the width base case, through the iteration (3.3) of [9], whose iterates stay in the box of that call because T clips to it; by Lemma 3.4 of [9] it makes at most $522240n + 1$ evaluations. □

Lemma 4.5 (arithmetic at a node)**.** *Assume Assumption A(b),(c). In the tree of Lemma 4.4, consider one node, and do not count the evaluations of* $\lfloor F \rfloor$ *nor the work done inside its children.*

(a) Every integer it computes has absolute value at most $\Xi := 2\max(\|A\|_\infty, \|A + h2^d\mathbf{1}\|_\infty) + 2h2^d$*. The only other integers it handles are the oracle values* $\lfloor F_i(t) \rfloor$ *returned in the width base case; these are not bounded by* Ξ *in general, and the node does nothing with them except compare them with box ends.*

(b) Every mass it handles is $1/n$ *times a product of at most* B_0 *constants from* $\{\frac{4}{3}, \frac{4}{5}, \frac{8}{5}\}$*; kept as an unreduced fraction it has a numerator of at most* $3B_0 + 1$ *bits and a denominator of at most* $\log_2 n + 3B_0 + 1$ *bits.*

(c) It performs at most $c_0\, n\, (522240n + 1)$ *operations, for an absolute constant* c_0*, each of which is an addition, subtraction, comparison, or halving of integers bounded as in (a), a comparison of an oracle value* $\lfloor F_i(t) \rfloor$ *with an integer bounded as in (a), a multiplication of a mass by a constant from* $\{\frac{4}{3}, \frac{4}{5}, \frac{8}{5}\}$*, a multiplication of two masses, or a comparison of a mass or of a product of two masses with* 1*.*

Proof. We go through the procedure of Appendix A. Let the node be the call $\mathcal{L}(A', W', a', b')$, with $u' = A' + W'\mathbf{1}$; by Lemma 3.4 of [9] its box lies in the root box.

(a) The integers handled are: the box ends A', u'; the scalars W', $W'/2$, P, M, q, $q/2$, p, which are at most $W' \leq h2^d$ (and are integers in a nonbase call, where the exponent d' of W' satisfies $d' \geq 12$; in a base call only $W'/2$ is used, in a comparison); the centre c; the pivots k, k^{new}, which stay between $c - M\mathbf{1}$ and $c + M\mathbf{1}$; the lower ends $k - (r/2)\mathbf{1}$ of the boxes of centered calls, which lie in the box of the node by Lemma 3.4 of [9]; and, in the width base case, the iterates $t^m \in [A', u']$ and the vectors $\lfloor F(t^m) \rfloor$ returned by the oracle, which are only compared with A' and u'. All but the last are bounded by $\Xi/2$, and the only other integers formed are the doubles $2t_i^*$ and $2A_i' + W'$ in the output test of the width base case; the oracle values are covered by Proposition 5.5(d) in our application and are not operated on otherwise.

(b) The masses of a child differ from those of its parent only for the children of the two passes (one of the two vectors is multiplied by $\frac{4}{3}$) and for the continuation (each a_i, b_i is multiplied by $\frac{4}{5}$ or $\frac{8}{5}$); the provisional call has the masses of its parent. By Lemma 3.5 of [9] the first two kinds of children are exactly the B-decreasing ones, and a root-to-node path has at most B_0 of them, since B is a nonnegative integer that never increases along the path and equals B_0 at the root. So each a_i', b_i' is $\frac{1}{n}$ times a product of at most B_0 of the three constants: numerator at most $8^{B_0} = 2^{3B_0}$, which has $3B_0 + 1$ bits, and denominator at most $n\, 5^{B_0} < 2^{\log_2 n + 3B_0}$.

(c) *Mass base test* (made by every node): n products $a_i'b_i'$, n comparisons of them with 1, and, if the case applies, n comparisons of b_i' with 1. *Width base case*: each evaluation of T in

(3.3) of [9] takes, besides the oracle call, $2n$ comparisons for the clipping and n comparisons for the test $T(t) = t$; there are at most $522240n + 1$ evaluations (Lemma 3.4 of [9]); the output takes n comparisons of $2t_i^*$ with $2A'_i + W'$. *Nonbase call*: computing c and M, q, p, P takes $O(n)$ operations. One iteration of a pass forms the input of a centered call (n subtractions for the lower end, n multiplications of masses by $\frac{4}{3}$), and after the call returns computes k^{new} (n additions or subtractions and n comparisons) and tests $k^{\text{new}} = k$ (n comparisons): $O(n)$ operations. By Lemma 3.4 of [9] each pass has at most $1024n + 1$ iterations, so the two passes take $O(n^2)$ operations. The sets I_- and I_+ take n comparisons each. The provisional call takes n subtractions to form its input; the reweighting takes $2n$ multiplications of masses by constants; the overrides take $O(n)$ operations. Altogether a nonbase node performs $O(n^2)$ operations and a width base node $O(n(522240n + 1))$. □

5 An exact, monotone, translation-equivariant surrogate of log-sum-exp

For an integer $\zeta \geq 1$ let $P_\zeta(y) = \sum_{i=0}^{\zeta} y^i/i!$ and

$$E(x) = \begin{cases} P_\zeta(x), & x \geq 0, \\ 1/P_\zeta(-x), & x < 0. \end{cases} \tag{4}$$

Lemma 5.1. *(a) E is continuous and strictly increasing on $\mathbb{R}$, $E(0) = 1$, $E(x) \to 0$ as $x \to -\infty$ and $E(x) \to \infty$ as $x \to \infty$. Moreover $E(x) \leq e^x$ for $x \geq 0$ and $E(x) \geq e^x$ for $x \leq 0$.*

(b) Let $K_0 \geq 0$ and $0 < \eta_0 \leq 1$. If $\zeta + 1 \geq \max(2eK_0, \log_2(2/\eta_0))$ then $P_\zeta(y) \geq (1 - \eta_0)e^y$ for all $0 \leq y \leq K_0$.

Proof. (a) P_ζ is continuous, strictly increasing on $[0, \infty)$, $P_\zeta(0) = 1$, $P_\zeta \leq \exp$ there and $P_\zeta(y) \to \infty$. (b) For $0 \leq y \leq K_0$ the ratio of consecutive terms of $\sum_{i>\zeta} y^i/i!$ is at most $y/(\zeta + 2) \leq \frac{1}{2}$, so $e^y - P_\zeta(y) \leq 2y^{\zeta+1}/(\zeta + 1)! \leq 2\big(ey/(\zeta + 1)\big)^{\zeta+1} \leq 2 \cdot 2^{-(\zeta+1)} \leq \eta_0 \leq \eta_0 e^y$, using $k! \geq (k/e)^k$. □

Fix a People state p with outgoing arcs (p, j), $j \in D_p \subseteq D$, and weights $m_{pj} > 0$. For $t \in \mathbb{R}^D$ and $s \in \mathbb{R}$ put

$$\Phi_p(t, s) = \sum_{j \in D_p} m_{pj}\, E(t_j - s).$$

By Lemma 5.1(a), $s \mapsto \Phi_p(t, s)$ is continuous and strictly decreasing from $+\infty$ to 0. Let $g_p(t)$ be the unique s with $\Phi_p(t, s) = 1$.

Lemma 5.2 (exactness)**.** *For all $t, t' \in \mathbb{R}^D$, $s, c \in \mathbb{R}$:*

(a) $g_p(t) \geq s$ if and only if $\Phi_p(t, s) \geq 1$;

(b) $t \leq t'$ implies $g_p(t) \leq g_p(t')$;

(c) $g_p(t + c\mathbf{1}) = g_p(t) + c$.

These hold for every continuous strictly increasing $E : \mathbb{R} \to (0, \infty)$ onto $(0, \infty)$.

Proof. (a) $\Phi_p(t, \cdot)$ is strictly decreasing and equals 1 at $g_p(t)$. (b) If $t \leq t'$ then $\Phi_p(t', s) \geq \Phi_p(t, s)$ for all s; with $s = g_p(t)$ and (a), $g_p(t') \geq g_p(t)$. (c) $\Phi_p(t + c\mathbf{1}, s + c) = \Phi_p(t, s)$. □

Lemma 5.3 (accuracy)**.** *Let $0 < \varepsilon \leq 1$, let $m_\Sigma \geq \sum_{j \in D_p} m_{pj}$ and $0 < m_{\min} \leq m_{pj}$ for all j, and let*

$$K_0 \geq \max\big(\ln(4m_\Sigma/\varepsilon),\ \ln(2/m_{\min})\big), \qquad \eta_0 = \varepsilon/4, \qquad \zeta + 1 \geq \max(2eK_0, \log_2(2/\eta_0)).$$

Then $|g_p(t) - \ell_p(t)| \leq \varepsilon$ for all *$t \in \mathbb{R}^D$.*

Proof. The hypotheses imply $K_0 \geq 0$ and $\zeta \geq 2$ (in particular $\zeta \geq 1$, as (4) requires), so that Lemma 5.1(b) applies. Indeed, D_p is nonempty, so $m_\Sigma \geq m_{\min}$: if $m_{\min} \leq 2$ then $\ln(2/m_{\min}) \geq 0$, and otherwise $4m_\Sigma/\varepsilon > 8$ and $\ln(4m_\Sigma/\varepsilon) > 0$; and $\zeta + 1 \geq \log_2(2/\eta_0) = \log_2(8/\varepsilon) \geq 3$.

Fix t and write $x_j = t_j - s$. By Lemma 5.1, for every real x: $E(x) \leq e^x$ if $x \geq 0$; $E(x) \leq e^x/(1-\eta_0)$ if $-K_0 \leq x < 0$; and $E(x) \leq 1/P_\zeta(K_0) \leq e^{-K_0}/(1-\eta_0)$ if $x < -K_0$. Also $E(x) \geq e^x$ if $x \leq 0$; $E(x) \geq (1-\eta_0)e^x$ if $0 \leq x \leq K_0$; and $E(x) \geq P_\zeta(K_0) \geq (1-\eta_0)e^{K_0}$ if $x > K_0$.

Upper bound. Take $s = \ell_p(t) + \varepsilon$, so that $\sum_j m_{pj} e^{x_j} = e^{-\varepsilon}$. Then

$$\Phi_p(t,s) \leq \frac{\sum_j m_{pj} e^{x_j} + m_\Sigma e^{-K_0}}{1-\eta_0} \leq \frac{e^{-\varepsilon} + \varepsilon/4}{1-\varepsilon/4} \leq \frac{1 - \varepsilon/2 + \varepsilon/4}{1-\varepsilon/4} = 1,$$

using $e^{-\varepsilon} \leq 1 - \varepsilon + \varepsilon^2/2 \leq 1 - \varepsilon/2$. Hence $g_p(t) \leq s$.

Lower bound. Take $s = \ell_p(t) - \varepsilon$, so that $\sum_j m_{pj} e^{x_j} = e^{\varepsilon}$. If some $x_j > K_0$ then $\Phi_p(t,s) \geq m_{\min}(1-\eta_0)e^{K_0} \geq 2(1-\eta_0) \geq 1$. Otherwise $\Phi_p(t,s) \geq (1-\eta_0)e^{\varepsilon} \geq (1-\varepsilon/4)(1+\varepsilon) \geq 1$. Hence $g_p(t) \geq s$ by Lemma 5.2(a). □

The point of the two extreme ranges is that E decays and grows only polynomially where exp decays and grows exponentially; the proof shows that this is harmless: a coordinate far below s contributes a negligible amount under both E and exp, and a coordinate far above s makes both sums exceed 1.

Definition 5.4 (surrogate operators)**.** With g_p as above for every People state p, let

$$\tilde{f}_d(t) = \min_{(d,\vartheta)\in E} \max_{(\vartheta,p)\in E} g_p(t) \qquad (t \in \mathbb{R}^D,\ d \in D),$$

and for a rational $\delta > 0$, an integer $N \geq 2$ and $\gamma = 1 - 1/N$ let $G(t) = \gamma\, \tilde{f}(\delta t)/\delta$.

Proposition 5.5. *(a) $\tilde{f}$ is monotone and $\tilde{f}(t + c\mathbf{1}) = \tilde{f}(t) + c\mathbf{1}$ for all $c \in \mathbb{R}$; consequently G satisfies (3) exactly.*

(b) Under the hypotheses of Lemma 5.3 for every p, $\|\tilde{f}(t) - f(t)\|_\infty \leq \varepsilon$ for all $t \in \mathbb{R}^D$.

(c) For $t \in \mathbb{Z}^D$, $k \in \mathbb{Z}$ and $d \in D$,

$$G_d(t) \geq k \iff \forall (d,\vartheta) \in E\ \ \exists (\vartheta, p) \in E : \quad \sum_{j \in D_p} m_{pj}\, E\Big(\delta\big(t_j - \tfrac{kN}{N-1}\big)\Big) \geq 1.$$

(d) Under the hypotheses of (b), for the normalised weights of Section 2 and all $t \in \mathbb{R}^D$, $\gamma(\min_j t_j - (\Omega+\varepsilon)/\delta) \leq G_d(t) \leq \gamma(\max_j t_j + (\Omega+\varepsilon)/\delta)$.

Proof. (a) Minima and maxima of maps with the two properties of Lemma 5.2(b),(c) have them too; then $G(t+c\mathbf{1}) = \gamma(\tilde{f}(\delta t) + \delta c\mathbf{1})/\delta = G(t) + \gamma c\mathbf{1}$. (b) Minima and maxima are nonexpansive for the sup norm. (c) $G_d(t) \geq k$ iff $\tilde{f}_d(\delta t) \geq \delta k/\gamma$ iff for all ϑ there is p with $g_p(\delta t) \geq \delta k N/(N-1)$; apply Lemma 5.2(a). (d) $\min_j y_j + \ln m_{\min} \leq \ell_p(y) \leq \max_j y_j + \ln \sum_j m_{pj}$, and $|\ln m_{\min}| \leq \Omega$, $\ln \sum_j m_{pj} \leq \ln(n a_{\max}) \leq \Omega$; use (b). □

Lemma 5.6 (cost of one evaluation)**.** *Assume the hypotheses of Proposition 5.5(b). Let $t \in \mathbb{Z}^D$ with $\|t\|_\infty \leq 2^\beta$, and let $\delta = 1/\Delta$ with Δ a positive integer. Then $\lfloor G(t) \rfloor$ can be computed in a number of steps polynomial in L, β, $\log N$, $\log \Delta$, ζ and an integer upper bound $\widehat{\Omega}$ of Ω.*

Proof. By Proposition 5.5(d) the integer $\lfloor G_d(t) \rfloor$ lies in an interval with integer end points of $O(\beta + \log\Delta + \log\widehat{\Omega})$ bits, and by (c) and monotonicity in k it is found by binary search, each step being the test in (c). In that test each argument $z = \delta(t_j - kN/(N-1))$ is a rational whose numerator and denominator have $O(\beta + \log N + \log\Delta + \log\widehat{\Omega})$ bits; $P_\zeta(|z|)$ is obtained by Horner's rule with ζ multiplications and additions of rationals whose sizes stay $O(\zeta(\mathrm{size}(z) + \log\zeta))$; then $E(z)$ is $P_\zeta(|z|)$ or its inverse, and the sum over at most n arcs is compared with 1. There are at most L arcs in total. □

6 A quantitative discount bound

In this section M is a nonnegative real $n \times n$ matrix every row of which has a positive entry, the positive entries lie in $[m_{\min}, m_{\max}]$, and Ω is any real with

$$\Omega \geq \ln\bigl(n \max(m_{\max}, 1/m_{\min}, 1)\bigr). \tag{5}$$

Thus $|\ln m| \leq \Omega$ for every positive entry m, $\ln(nm_{\max}) \leq \Omega$ and $\ln(nm_{\max}/m_{\min}) \leq 2\Omega$.

Classes. Write $i \to j$ if $M_{ij} > 0$; j is *accessible* from i if there is a path from i to j (possibly of length 0). The *classes* are the strongly connected components. A class C is *nontrivial* if it contains an arc (possibly a loop); then every $i \in C$ has an arc into C, the submatrix M_{CC} is irreducible, and by the Perron–Frobenius theorem there are $\rho_C > 0$ and a vector $U = U^C \in \mathbb{R}^C_{>0}$ with $M_{CC}U = \rho_C U$. We normalise $\min_i U_i = 1$. A class C' is *below* C if $C' \neq C$ and C' is accessible from C; the *height* of C is the length of the longest chain of classes below it. Since every row is nonzero, the classes of height 0 are nontrivial and from every state some nontrivial class is accessible. Put

$$\lambda_i = \max\{\ln \rho_C : \ C \text{ nontrivial and accessible from } i\}.$$

Lemma 6.1. *(a) $m_{\min} \leq \rho_C \leq nm_{\max}$, so $|\ln \rho_C| \leq \Omega$ and $|\lambda_i| \leq \Omega$.*

(b) $0 \leq \ln U_i^C \leq \Upsilon_n := 2(n-1)\Omega$ for all $i \in C$.

(c) If $i \to j$ then $\lambda_j \leq \lambda_i$.

(d) $\lim_{k\to\infty} \frac{1}{k} \ln(M^k \mathbf{1})_i = \lambda_i$ for every i. In particular (apply this to the matrix M_{CC}) ρ_C does not depend on the choice of the eigenvector.

Proof. (a) At an index $i \in C$ where U_i is minimal, $\rho_C U_i = \sum_{j\in C} M_{ij}U_j \geq U_i \sum_{j\in C} M_{ij} \geq U_i m_{\min}$; at an index where U_i is maximal, $\rho_C U_i \leq U_i n m_{\max}$. (b) For an arc $i \to j$ inside C, $\rho_C U_i \geq M_{ij}U_j$, so $U_j/U_i \leq \rho_C/m_{\min} \leq nm_{\max}/m_{\min} \leq e^{2\Omega}$; any two states of C are joined inside C by a path with at most $n-1$ arcs, and $\min U = 1$. (c) Every class accessible from j is accessible from i.

(d) *Lower bound.* Let C be nontrivial, accessible from i, with $\ln \rho_C = \lambda_i$, and take a path from i to some $j \in C$ with $s \leq n-1$ arcs. For $k \geq s$, $(M^k\mathbf{1})_i \geq m_{\min}^s (M^{k-s}\mathbf{1})_j \geq m_{\min}^s (M_{CC}^{k-s}U)_j / \max U = m_{\min}^s \rho_C^{k-s} U_j / \max U$. *Upper bound.* Fix $\epsilon > 0$. We construct $u \in \mathbb{R}^n_{>0}$ with $(Mu)_i \leq e^{\lambda_i+\epsilon}u_i$ for all i, class by class in order of increasing height. Let C be a class and suppose u is defined on all classes below it; put $S = \max_{i\in C} \sum_{j\notin C} M_{ij}u_j$ (the sum runs over states in classes below C). If C is nontrivial put $u_i = \kappa U_i$ on C; then $(Mu)_i = \kappa\rho_C U_i + \sum_{j\notin C} M_{ij}u_j \leq \kappa e^{\lambda_i}U_i + S \leq e^{\lambda_i+\epsilon}\kappa U_i$ as soon as $\kappa > 0$ and $\kappa \geq S/((e^\epsilon - 1)e^{-\Omega})$ (if $S = 0$ any $\kappa > 0$ will do). If $C = \{i\}$ is trivial put $u_i = \kappa$ with $\kappa > 0$ and $\kappa \geq Se^{\Omega}$, so that $(Mu)_i \leq S \leq e^{\lambda_i+\epsilon}u_i$. Now $(M^k u)_i \leq e^{k(\lambda_i+\epsilon)}u_i$ for all $k \geq 0$ by induction: $(M^{k+1}u)_i = \sum_j M_{ij}(M^k u)_j \leq \sum_j M_{ij}e^{k(\lambda_j+\epsilon)}u_j \leq e^{k(\lambda_i+\epsilon)}(Mu)_i$, using (c). Hence $(M^k\mathbf{1})_i \leq (M^k u)_i / \min u \leq e^{k(\lambda_i+\epsilon)}u_i / \min u$. □

Let $N \geq 2$ be an integer, $\theta = 1/N$, $\gamma = 1 - \theta$. The map $\Psi_M(t) = \gamma \ln(Me^t)$ (coordinatewise logarithm and exponential) satisfies (3); let v be its fixed point.

Theorem 6.2 (discount bound)**.** *Let M be a nonnegative $n \times n$ matrix every row of which has a positive entry, with positive entries in $[m_{\min}, m_{\max}]$, let Ω satisfy (5), and let λ_i and v be as above. If $N \geq 2n$ then, for every i,*

$$|\theta v_i - \lambda_i| \leq \theta R, \qquad R := 3n\bigl(2\ln N + 4n\Omega + 2\bigr).$$

Proof. Lower bound. Fix i, a nontrivial class C accessible from i with $\ln \rho_C = \lambda := \lambda_i$, and $U = U^C$. Dropping nonnegative terms, $v_j \ge \gamma \ln\bigl(\sum_{j' \in C} M_{jj'} e^{v_{j'}}\bigr)$ for $j \in C$, that is, $v_C \ge \Psi_C(v_C)$ with $\Psi_C(t) = \gamma \ln(M_{CC} e^t)$ on $\mathbb{R}^C$, a map satisfying (3). Let $w_j = c + \ln U_j$ with $c = (\gamma\lambda - \theta\Upsilon_n)/\theta$. Then $\Psi_C(w)_j = \gamma(\lambda + c + \ln U_j) \ge c + \ln U_j$, because this is equivalent to $\gamma\lambda - \theta c \ge \theta \ln U_j$ and $\gamma\lambda - \theta c = \theta\Upsilon_n$. By Lemma 4.1, $w \le \mathrm{fix}(\Psi_C) \le v_C$. Hence for $j \in C$

$$\theta v_j \ge \theta c = \gamma\lambda - \theta\Upsilon_n \ge \lambda - \theta(\Omega + \Upsilon_n) =: z, \qquad |z| \le 2\Omega + \Upsilon_n.$$

Take a path $i = i_0 \to i_1 \to \cdots \to i_k$ with $i_k \in C$ and $k \le n-1$. From $v_{i_s} \ge \gamma(\ln m_{\min} + v_{i_{s+1}}) \ge \gamma v_{i_{s+1}} - \Omega$ we get by induction $v_{i_0} \ge \gamma^k v_{i_k} - k\Omega$. Therefore, since $\gamma^k > 0$ and $\theta v_{i_k} \ge z$ (no sign condition on z is needed here), and using $1 - \gamma^k \le k\theta$,

$$\theta v_i \ge \gamma^k z - k\theta\Omega \ge z - k\theta|z| - k\theta\Omega \ge \lambda - \theta\bigl(\Omega + \Upsilon_n + n(2\Omega + \Upsilon_n) + n\Omega\bigr) \ge \lambda - 5n^2\Omega\,\theta.$$

Upper bound. Define $R_0 = \Omega + \Upsilon_n$ and $R_{g+1} = (1 + 2\theta)(R_g + \Omega + \ln N + 1) + \Omega + \Upsilon_n$; the sequence is increasing. We prove by induction on g: *for every i in a class of height g, $\theta v_i \le \lambda_i + \theta R_g$.*

Height 0. The class C is nontrivial, no arc leaves it, and $\lambda := \lambda_i = \ln \rho_C$ on C. Let $w = c + \ln U$ with $c = \gamma\lambda/\theta$. Then $\Psi_C(w)_j = \gamma(\lambda + c + \ln U_j) = c + \gamma \ln U_j \le w_j$, so $v_C \le w$ and $\theta v_j \le \gamma\lambda + \theta\Upsilon_n \le \lambda + \theta(\Omega + \Upsilon_n)$.

Induction step. Let C have height $g+1$ and put $R' = R_g$. Every state $j \notin C$ with $M_{ij} > 0$ for some $i \in C$ lies in a class of height at most g, so by induction and Lemma 6.1(c), with $\lambda := \lambda_i$ (which is constant on C),

$$v_j \le \lambda_j/\theta + R' \le \lambda/\theta + R' =: b. \tag{6}$$

Trivial class $C = \{i\}$. All arcs from i leave C, so $v_i = \gamma \ln \sum_j M_{ij} e^{v_j} \le \gamma(\Omega + b)$, hence $\theta v_i \le \gamma(\lambda + \theta(\Omega + R')) \le \lambda + \theta(2\Omega + R') \le \lambda + \theta R_{g+1}$.

Nontrivial class C. Here $\lambda \ge \ln \rho_C$. On $\mathbb{R}^C$ consider

$$\Psi(t)_i = \gamma \ln\Bigl(\sum_{j \in C} M_{ij} e^{t_j} + \sum_{j \notin C} M_{ij} e^{v_j}\Bigr), \qquad i \in C.$$

Ψ is monotone, $\Psi(t + c\mathbf{1}) \le \Psi(t) + \gamma c\mathbf{1}$ for $c \ge 0$, and $\Psi(v_C) = v_C$; by the remark after Lemma 4.1, a vector w with $\Psi(w) \le w$ satisfies $v_C \le w$. Let

$$\eta = \frac{\theta}{\gamma}\,(R' + \Omega + \ln N + 1), \qquad c = \frac{\gamma}{\theta}(\lambda + \eta), \qquad w_i = c + \ln U_i \ (i \in C).$$

We estimate the argument of the logarithm in $\Psi(w)_i$. First, $\sum_{j \in C} M_{ij} e^{w_j} = e^c \rho_C U_i \le e^{\lambda + c} U_i$. Second, by (6) and $\ln(n m_{\max}) \le \Omega$, $\sum_{j \notin C} M_{ij} e^{v_j} \le e^{\Omega + b}$, and

$$b - c = \frac{\lambda}{\theta} + R' - \frac{\gamma\lambda}{\theta} - \frac{\gamma\eta}{\theta} = \lambda + R' - (R' + \Omega + \ln N + 1) = \lambda - \Omega - \ln N - 1,$$

so $e^{\Omega + b} = e^{\lambda + c}/(eN) \le e^{\lambda + c} U_i/(eN)$ because $U_i \ge 1$. Therefore

$$\Psi(w)_i \le \gamma\Bigl(\lambda + c + \ln U_i + \ln\bigl(1 + \tfrac{1}{eN}\bigr)\Bigr) \le \gamma(w_i + \lambda + \theta/e).$$

On the other hand $w_i/\gamma = w_i + \frac{\theta}{\gamma}(c + \ln U_i) \ge w_i + \frac{\theta}{\gamma} c = w_i + \lambda + \eta$, since $\ln U_i \ge 0$. As $\eta \ge \theta \ge \theta/e$ (because $R' + \Omega + \ln N + 1 \ge 1$ and $\gamma < 1$), we obtain $\Psi(w)_i \le \gamma \cdot w_i/\gamma = w_i$. Hence $v_C \le w$ and, for $i \in C$,

$$\theta v_i \le \theta c + \theta \ln U_i = \gamma(\lambda + \eta) + \theta \ln U_i \le \lambda + \theta\Omega + \eta + \theta\Upsilon_n,$$

where we used $\gamma(\lambda+\eta) = \lambda+\eta-\theta(\lambda+\eta) \le \lambda+\eta+\theta|\lambda|$. Finally $\theta/\gamma = \theta(1+\theta/\gamma) \le \theta(1+2\theta)$ since $\gamma \ge \frac{1}{2}$, so $\eta \le \theta(1+2\theta)(R'+\Omega+\ln N+1)$ and $\theta v_i \le \lambda + \theta R_{g+1}$.

Solving the recurrence. Let $a = 1+2\theta$ and $Q = 2(\Omega+\ln N+1)+\Omega+\Upsilon_n \ge a(\Omega+\ln N+1)+\Omega+\Upsilon_n$. Then $R_{g+1} \le aR_g + Q$, so $R_g \le a^g(R_0+gQ)$. The height is at most $n-1$ and $a^n \le e^{2n/N} \le e$ because $N \ge 2n$. Hence

$$R_g \le e(\Omega+\Upsilon_n+n(2\ln N+2+3\Omega+\Upsilon_n)) \le e\,n\,(2\ln N+2+4\Omega+2\Upsilon_n) \le 3n(2\ln N+4n\Omega+2),$$

as $4\Omega+2\Upsilon_n = 4n\Omega$. The constant $5n^2\Omega$ of the lower bound is smaller than R. □

Remark 6.3. In the reducible case R cannot be taken independent of N: for the 2×2 matrix with $M_{11} = M_{22} = M_{12} = 1$, $M_{21} = 0$ one has $\lambda_1 = \lambda_2 = 0$, $v_2 = 0$, and $X = e^{v_1}$ solves $X^{N/(N-1)} = X+1$, so that $v_1 = \ln N - \ln\ln N + o(1)$ as $N \to \infty$.

7 From matrices to the game, and separation

We return to the normalised game of Section 2; $\Omega = \ln(n\max(a_{\max}, q))$ satisfies (5) for every matrix $M^{\sigma\tau}$. Write $\lambda_d^{\sigma\tau}$ for the numbers λ_d of the matrix $M^{\sigma\tau}$.

Lemma 7.1. *(a) $\chi_d = \min_\sigma \max_\tau \lambda_d^{\sigma\tau}$, the minimum and maximum being over policies. (This part does not use the normalisation.)*

(b) Let $\xi : \mathbb{R}^D \to \mathbb{R}^D$ be of the form $\xi_d(t) = \min_{(d,\vartheta)\in E} \max_{(\vartheta,p)\in E} \xi_p(t)$, where each $\xi_p : \mathbb{R}^D \to \mathbb{R}$ is monotone and satisfies $\xi_p(t+c\mathbf{1}) = \xi_p(t)+c$, and for policies σ,τ put $\xi_d^{\sigma\tau} = \xi_{\tau(\sigma(d))}$. Let y and $y^{\sigma\tau}$ be the fixed points of $\gamma\xi$ and $\gamma\xi^{\sigma\tau}$. Then $y = \min_\sigma \max_\tau y^{\sigma\tau}$ coordinatewise.

(c) If $N \ge 2n$ and v is the fixed point of γf, then $|\theta v_d - \chi_d| \le \theta R$ for all d, with R as in Theorem 6.2.

(d) If $\|\tilde f - f\|_\infty \le \varepsilon$ and y is the fixed point of $\gamma\tilde f$, then $\theta\|y-v\|_\infty \le \varepsilon$.

Proof. (a) For policies, $R_d^k(\sigma,\tau) = ((M^{\sigma\tau})^k\mathbf{1})_d$, so $R_d^\infty(\sigma,\tau) = \exp(\lambda_d^{\sigma\tau})$ by Lemma 6.1(d). By [2, Theorem 10] there are policies σ^*,τ^* with $R_d^\infty(\sigma,\tau^*) \ge V_d^\infty \ge R_d^\infty(\sigma^*,\tau)$ for all strategies, in particular all policies, σ,τ. Hence $\max_\tau \lambda_d^{\sigma^*\tau} \le \chi_d$ and $\max_\tau \lambda_d^{\sigma\tau} \ge \lambda_d^{\sigma\tau^*} \ge \chi_d$ for every σ.

(b) All maps involved satisfy (3). Choose σ^* with $\xi_d(y) = \max_{(\sigma^*(d),p)\in E} \xi_p(y)$ for all d. For every τ, $\gamma\xi^{\sigma^*\tau}(y) \le \gamma\xi(y) = y$, so $y^{\sigma^*\tau} \le y$ by Lemma 4.1. For an arbitrary σ choose τ_σ with $\xi_{\tau_\sigma(\sigma(d))}(y) = \max_{(\sigma(d),p)\in E} \xi_p(y) \ge \xi_d(y)$; then $\gamma\xi^{\sigma\tau_\sigma}(y) \ge y$ and $y^{\sigma\tau_\sigma} \ge y$. Thus $\max_\tau y^{\sigma^*\tau} \le y \le \max_\tau y^{\sigma\tau}$ for all σ.

(c) Apply (b) to $\xi_p = \ell_p$: $v = \min_\sigma \max_\tau v^{\sigma\tau}$, where $v^{\sigma\tau}$ is the fixed point of $t \mapsto \gamma\ln(M^{\sigma\tau}e^t)$. By Theorem 6.2, $|\theta v_d^{\sigma\tau} - \lambda_d^{\sigma\tau}| \le \theta R$ for all pairs, and min max of families that are termwise within θR are within θR; conclude with (a).

(d) $\|y-v\| = \gamma\|\tilde f(y) - f(v)\| \le \gamma\|\tilde f(y) - f(y)\| + \gamma\|f(y)-f(v)\| \le \gamma\varepsilon + \gamma\|y-v\|$, so $\|y-v\| \le \gamma\varepsilon/\theta$. □

Lemma 7.2 (separation at threshold 1)**.** *Let $H = n\,(q+2na_{\max})^n$. For all policies σ,τ and all d, either $\lambda_d^{\sigma\tau} = 0$ or $|\lambda_d^{\sigma\tau}| \ge 1/H$. Consequently, for every d, either $\chi_d = 0$ or $|\chi_d| \ge 1/H$.*

Proof. Let C be a nontrivial class of $M = M^{\sigma\tau}$ and $\rho = \rho_C$. Then $\widehat M := qM_{CC}$ is a $k\times k$ matrix with nonnegative integer entries at most $a_{\max}$, $k \le n$, and $\alpha := q\rho$ is an eigenvalue of $\widehat M$. Suppose $\alpha \ne q$. Let ϕ be the characteristic polynomial of $\widehat M$ (monic, integer coefficients) and write $\phi(z) = (z-q)^\nu\psi(z)$ with $\psi(q) \ne 0$; ψ is monic with integer coefficients (division by a monic integer polynomial), $\psi(\alpha) = 0$, and $\psi(q)$ is a nonzero integer. All roots of ψ are eigenvalues of $\widehat M$ and have modulus at most the maximal row sum, $na_{\max}$. Write $\psi(z) = \prod_r (z-r)$, the product being over the complex roots r of ψ counted with multiplicity; one of them is α, there are at

most n of them, and $|q-r| \le q + na_{\max}$ for each. Hence $1 \le |\psi(q)| \le |q-\alpha|\,(q+na_{\max})^{n-1}$. Finally, since $0 < \alpha \le na_{\max}$, $|\ln(\alpha/q)| \ge |\alpha - q|/\max(\alpha, q) \ge (q+na_{\max})^{-n} \ge 1/H$. So $\ln \rho_C$ is 0 or at least $1/H$ in absolute value; $\lambda_d^{\sigma\tau}$ is one of these numbers and χ_d is one of the $\lambda_d^{\sigma\tau}$ by Lemma 7.1(a). □

The integer H of Lemma 7.2 satisfies $\log_2 H \le \log_2 n + n(L + 2 + \log_2 n) = O(L^2)$.

8 The algorithm

Parameters. Given the normalised game $(n,\, q,\, a_{\max})$, compute

$$H = n(q+2na_{\max})^n, \qquad \widehat{\Omega} = \text{the bit length of } n\max(a_{\max}, q) \quad (\widehat{\Omega} \ge \Omega),$$
$$N = 2^s \text{ with } s \text{ the least integer such that } 2^s \ge 2n \text{ and } 64H \cdot 3n(2s + 4n\widehat{\Omega} + 2) \le 2^s,$$
$$\varepsilon = \frac{1}{64H}, \qquad K_0 = \text{bitlength}(256Hna_{\max}) + \text{bitlength}(2q),$$
$$\zeta = \max\bigl(6K_0,\ \text{bitlength}(512H) + 1\bigr),$$
$$\delta = \frac{1}{2^{15}H}, \qquad K = 256(N+1), \qquad J = 2^{15}HN(\widehat{\Omega}+1) + N + 1.$$

Since $\ln N \le s$, the choice of N gives $\theta R \le 1/(64H)$ for R of Theorem 6.2 (with $\Omega \le \widehat{\Omega}$). The hypotheses of Lemma 5.3 hold with $m_\Sigma = na_{\max}$, $m_{\min} = 1/q$, $\eta_0 = 1/(256H)$: indeed $K_0 \ge \ln(256Hna_{\max}) = \ln(4m_\Sigma/\varepsilon)$, $K_0 \ge \ln(2q)$, $\zeta+1 \ge 6K_0 \ge 2eK_0$ and $\zeta+1 \ge \log_2(512H) = \log_2(2/\eta_0)$. The integers H, N, K, J and the denominators of ε and δ have $O(L^2)$ bits; ζ, K_0, $\widehat{\Omega}$, s are $O(L^2)$ in value. Consequently, when Lemma 5.6 is applied with these parameters at integer points of $O(L^2)$ bits, the size bound $O(\zeta(\text{size}(z) + \log\zeta))$ in its proof shows that each value $P_\zeta(|z|)$, and hence each $E(z)$, is a rational of $O(L^4)$ bits; the sum over at most $n \le L$ arcs that is compared with 1 therefore has $O(L^5)$ bits.

The role of each parameter is summarised in Table 1.

Parameter	Role	Used in
H	separation: $\chi_d = 0$ or $\lvert\chi_d\rvert \ge 1/H$	Lemma 7.2
$\widehat{\Omega}$	integer upper bound of Ω	Proposition 5.5(d), Theorem 6.2
$N = 2^s$	discount factor $\gamma = 1 - 1/N$, with $\theta R \le 1/(64H)$	Theorem 6.2, Lemma 7.1(c)
ε	accuracy of the surrogate, $\lVert\tilde{f} - f\rVert_\infty \le \varepsilon$	Lemma 5.3, Proposition 5.5(b)
ζ	degree of the polynomial P_ζ defining E	Equation (4), Lemma 5.3
K_0	range $[0, K_0]$ on which P_ζ approximates exp	Lemmas 5.1(b), 5.3
η_0	relative accuracy of P_ζ on $[0, K_0]$; $\eta_0 = \varepsilon/4$	Lemmas 5.1(b), 5.3
δ	scaling of the argument in G, with $\theta\delta K \le 1/(64H)$	Definition 5.4
K	width at which the loop stops	Proposition 4.3
J	the initial box starts at $-J\mathbf{1}$; $\lVert x\rVert_\infty + N + 1 \le J$	Proposition 4.3
d_0	exponent of the initial width $128 \cdot 2^{d_0} \ge 2J$	Proposition 4.3

Table 1: The parameters of the algorithm; η_0 occurs only in the analysis, and d_0 is computed in Step 3 of Algorithm 1.

Algorithm. The algorithm is stated as Algorithm 1. With E as in (4) for this ζ, and G as in Definition 5.4 for these δ, N, it runs the loop of Proposition 4.3 for G, J, K, each evaluation of $\lfloor G \rfloor$ being done as in Lemma 5.6. If $A \in \mathbb{Z}^D$ is the result, the output is $\{d \in D :\ A_d \ge -2^{13}N\}$.

Algorithm 1 (threshold problem for entropy games).

Input: an entropy game with positive rational weights and a rational $r > 0$, in binary.
Output: the set $\{d \in D : \ V_d^\infty \geq r\}$.

1. Normalise as in Section 2: compute q and the a_e, so that the weights are $m_e = a_e/q$ and the threshold is 1.
2. Compute H, $\widehat{\Omega}$, N, ε, K_0, ζ, δ, K, J as in the paragraph "Parameters". Let E be as in (4) for this ζ and G as in Definition 5.4 for these δ, N.
3. Set $A = -J\mathbf{1}$ and $W = 128 \cdot 2^{d_0}$, where d_0 is the least nonnegative integer with $128 \cdot 2^{d_0} \geq 2J$.
4. While $W > K$:
   (a) compute $\ell = \mathcal{L}(A, W, a, b)$ with $a_i = b_i = 1/n$ for the map $F = G$, each evaluation of $\lfloor G \rfloor$ required by $\mathcal{L}$ being done as in Lemma 5.6;
   (b) replace A_i by $A_i + (W - W^-)$ for every i with $\ell_i = +$, where W^- is the predecessor of W (defined before Proposition 4.3), and replace W by W^-.
5. Output $\{d \in D : \ A_d \geq -2^{13}N\}$.

Steps 3 and 4 are the loop of Proposition 4.3; $\mathcal{L}$ is the procedure of Appendix A.

Proof of Theorem 1.1. Correctness. By Proposition 5.5(a), G satisfies (3). Its fixed point is $x = y/\delta$ with y the fixed point of $\gamma\tilde{f}$. By Lemma 7.1(c),(d), Proposition 5.5(b) and the choice of parameters,

$$|\theta y_d - \chi_d| \leq \theta \|y - v\|_\infty + |\theta v_d - \chi_d| \leq \varepsilon + \theta R \leq \frac{1}{32H}.$$

Since $|\chi_d| \leq \Omega$ (Lemma 6.1(a) and Lemma 7.1(a)), $\|y\|_\infty \leq N(\widehat{\Omega} + 1)$, so $\|x\|_\infty + N + 1 \leq J$ and Proposition 4.3 applies: $A_d \leq x_d \leq A_d + K$. Multiplying by $\theta\delta$, $0 \leq \theta y_d - \theta\delta A_d \leq \theta\delta K = 256(N+1)/(2^{15}HN) \leq 1/(64H)$. Altogether

$$|\theta\delta A_d - \chi_d| \leq \frac{3}{64H} < \frac{1}{16H}.$$

The output test $A_d \geq -2^{13}N$ is the test $\theta\delta A_d \geq -1/(4H)$. If $\chi_d \geq 0$ then $\theta\delta A_d > -1/(16H)$ and d is output. If $\chi_d < 0$ then $\chi_d \leq -1/H$ by Lemma 7.2, so $\theta\delta A_d < -15/(16H)$ and d is not output. Recall that $\chi_d \geq 0$ means $V_d^\infty \geq r$ for the original weights.

Running time. Normalisation and the parameters are computed in time polynomial in L. We have $d_0 \leq \log_2 J = O(L^2)$. By Proposition 4.3 there are at most $128(d_0 + 1)$ calls of $\mathcal{L}$; by Lemma 4.4 each has at most $(\Lambda + 1)(B_0 + 1)(\Lambda s_*)^{B_0}$ nodes with $\Lambda = O(L^2)$, $s_* = O(L)$ and $B_0 = O(\log(L + 2))$, that is $2^{O((\log(L+2))^2)}$ nodes. All integer points at which $\lfloor G \rfloor$ is evaluated lie in $[-J, -J + 128 \cdot 2^{d_0}]^D$, so they have $O(L^2)$ bits; by Lemma 5.6 one evaluation of the vector $\lfloor G(t) \rfloor$ costs a polynomial in L. A node makes at most $522240n + 1$ such evaluations (Lemma 4.4) and at most $c_0 n(522240n + 1)$ other operations, on integers of $O(L^2)$ bits and on masses of $O(\log(L + 2))$ bits (Lemma 4.5, with $\Xi \leq 512 \cdot 2^{d_0}$, since every box of the loop lies in $[-J, -J + 128 \cdot 2^{d_0}]^D$ and $J \leq 64 \cdot 2^{d_0}$; the oracle values $\lfloor G_d(t) \rfloor$, which the node only compares with box ends, also have $O(L^2)$ bits by Proposition 5.5(d)).

Implementation on a Turing machine. The recursion is executed depth first with an explicit stack on a work tape. The frame of a call stores its input (A', h, d', a', b'), the step in progress, the current pivot, the number of iterations made in the current pass, the sets I_-, I_+, and the label vector last returned by a child: $O(n)$ integers of $O(L^2)$ bits, $O(n)$ masses of $O(\log(L + 2))$ bits and $O(n)$ further bits, that is $O(L^3)$ bits. By Lemma 4.4 the depth of the recursion is at most $\Lambda = O(L^2)$, so the stack never holds more than $O(L^5)$ bits. Each operation counted in Lemma 4.5, each push or pop of a frame, and each evaluation of $\lfloor G \rfloor$ (performed on separate

work tapes whose content is polynomial in L by Lemma 5.6) therefore costs a number of steps polynomial in L, including the head movements over the stack. The number of steps is at most

$$128(d_0+1)\cdot(\Lambda+1)(B_0+1)(\Lambda s_*)^{B_0}\cdot\mathrm{poly}(L)=2^{O((\log(L+2))^2)}.$$

□

Remark 8.1 (output)**.** The algorithm decides the sign of $\chi_d=\ln(V_d^\infty/r)$ including the case of equality; it does not compute the algebraic number V_d^∞ nor optimal policies. The separation bound of Lemma 7.2 concerns only the distance of $\ln V_d^\infty$ to the logarithm of the given rational.

9 Limits of the method and open problems

Proposition 9.1 (lower bound on the recursion tree)**.** *For every map F as in Equation (3.1) of [9] and every input $(A,h2^d,a,b)$ as in Equation (3.2) of [9], the recursion tree of the procedure $\mathcal{L}$ of Appendix A has at least $\binom{d-11+B}{B}$ nodes if $d\geq 11$, where $B=B(\mu(a,b))$. In the algorithm of Section 8 the first call of $\mathcal{L}$ alone has at least $\binom{d_0-11+B_0}{B_0}\geq\binom{n+B_0}{B_0}$ nodes, and $\binom{n+B_0}{B_0}\geq\big(n/(7\log_2 n)\big)^{5\log_2 n}$ when $n\geq 2^7$, whatever the game. (The bound is superpolynomial in the number n of Despot states; for bounded n it is polynomial in d_0, hence in L.)*

Proof. Let $\nu(d,B)$ be the minimum number of nodes over all inputs with these parameters. A call is a base call only if $B=0$ (mass base case) or $d<12$ (width base case). A nonbase call always makes the provisional call, which has parameters $(d-1,B)$, and the continuation, whose masses have all products multiplied by exactly ρ, so that its parameters are $(d,B-1)$. Hence $\nu(d,B)\geq 1+\nu(d-1,B)+\nu(d,B-1)\geq\nu(d-1,B)+\nu(d,B-1)$ for $d\geq 12$, $B\geq 1$, and $\nu\geq 1$ always; induction gives $\nu(d,B)\geq\binom{d-11+B}{B}$. In our algorithm the first call has $128\cdot 2^{d_0}\geq 2J$ with $J>2^{15}HN$, $H\geq 2^n$ and $N\geq 2$, so $d_0\geq n+11$ and the first call has at least $\binom{n+B_0}{B_0}\geq(n/B_0)^{B_0}$ nodes. Here $B_0=B(1/n^2)\geq\log_\rho n^2>5\log_2 n$ and $B_0\leq 2+6\log_2 n\leq 7\log_2 n$ for $n\geq 4$; for $n\geq 2^7$ the base $n/(7\log_2 n)$ exceeds 1, which gives the stated bound. □

So a polynomial bound cannot be obtained by a sharper analysis of the same procedure. Theorem 1.1 becomes unconditional once Assumption A is proved, for instance through a formal verification of Theorem 3.2 and Lemmas 3.4, 3.5 of [9] for the procedure of Appendix A; we do not address this here. The following are open.

1. Are entropy games solvable in polynomial time? Already the case of a fixed number of significant Tribune states (states in which Tribune has several moves) is stated as an open question in the conclusion of [3].

2. Can optimal policies be computed within the bound of Theorem 1.1?

3. Does the analogue of Theorem 6.2 hold for operators mixing weighted averages with log-sum-exp gates (risk-sensitive turn-based stochastic games)? The surrogate of Section 5 extends verbatim, since a weighted average is exactly monotone and translation-equivariant. For the entropic-risk objective with total reward, Baier, Chatterjee, Meggendorfer and Piribauer [5] show determinacy with memoryless deterministic optimal strategies and obtain decidability of the threshold problem through algebraic numbers for rational inputs. For the mean-payoff version with log-sum-exp gates and expectations, however, the analogue of the separation bound of Section 7 is a more serious obstacle than the discount bound. Already without players, an expectation over states i whose growth rates are $\ln w_i$, taken with probabilities p_i, has value $\sum_i p_i\ln w_i$, and comparing such a value with 0 for rational w_i and probabilities written in binary is the comparison of two products of powers of integers with exponents in binary. Etessami, Stewart and Yannakakis [7] show that this problem is decidable in

polynomial time if the Lang–Waldschmidt conjecture on linear forms in logarithms holds, and recall that it is decidable unconditionally when the number of terms is bounded (a fact they attribute to Shub); in general its complexity is, in their words, "intimately connected to deep conjectures and results in number theory". A separation bound with polynomially many bits for the mixed operators would therefore need a new idea or such a conjecture.

A The labelling procedure

We reproduce the procedure of [9, Section 3.2], whose text is used in Lemmas 4.4 and 4.5 and in Proposition 9.1. The wording is that of [9]; its displayed equations (3.4)–(3.9) are set inline, and text omitted between the two quoted passages is marked by [...].

"A *centered call* of even width r at an integer pivot k means the call $\mathcal{L}(k-(r/2)\mathbf{1}, r, a', b')$ with the indicated masses. All calls use the same map F. In a nonbase call set $c = A + \frac{D}{2}\mathbf{1}$, $M = \frac{D}{8}$, $q = \frac{D}{128}$, $p = \frac{q}{32} = \frac{D}{4096}$." [...]

"**Procedure $\mathcal{L}(A, D, a, b)$.** Execute the first applicable base case. If neither applies, execute Steps 3–6 in order.

1. **Mass base case.** If $a_i b_i > 1$ for every i, return $+$ at i when $b_i > 1$, and return $-$ otherwise.

2. **Width base case.** If $d < 12$, compute t^* by Equation (3.3). Return $+$ at i when $t_i^* > A_i + D/2$, and return $-$ otherwise.

3. **Downward pass.** Initialize $k = c$. Make a centered width-q call at k with masses $(a, \frac{4}{3}b)$, and use its labels to update all coordinates simultaneously by $k_i^{\mathrm{new}} = \max\{c_i - M, k_i - p\}$ if the child labels i by $-$, and $k_i^{\mathrm{new}} = k_i$ if the child labels i by $+$. Replace k by k^{new} and repeat until an update leaves the pivot unchanged. Denote this last pivot by $k^{\downarrow}$ and set $I_- = \{i : k_i^{\downarrow} < c_i - P\}$.

4. **Upward pass.** Start at $k = k^{\downarrow}$. Make a centered width-q call at k with masses $(\frac{4}{3}a, b)$, and update simultaneously by $k_i^{\mathrm{new}} = \min\{c_i + M, k_i + p\}$ if the child labels i by $+$, and $k_i^{\mathrm{new}} = k_i$ if the child labels i by $-$. Replace k by k^{new} and repeat until an unchanged update. Denote the last pivot by k^f and set $I_+ = \{i : k_i^f > c_i + P\}$.

5. **Provisional labels and new masses.** Make a centered width-$D/2$ call at k^f with masses (a, b). For its output ℓ^0, define $(\bar{a}_i, \bar{b}_i) = (\frac{4}{5}a_i, \frac{8}{5}b_i)$ if $\ell_i^0 = +$ and $(\bar{a}_i, \bar{b}_i) = (\frac{8}{5}a_i, \frac{4}{5}b_i)$ if $\ell_i^0 = -$.

6. **Continuation and overrides.** Call $\mathcal{L}(A, D, \bar{a}, \bar{b})$. In its output replace the signs on I_- by $-$ and those on $I_+ \setminus I_-$ by $+$. Return this label vector."

Equation (3.3) of [9] is the iteration $t^{m+1} = T(t^m)$, $T(t) = \min\{u, \max\{A, \lfloor F(t) \rfloor\}\}$, started at $t^0 = u$ and stopped at the first t^* with $T(t^*) = t^*$.

B Numerical checks

The proofs above do not depend on computation. As a safeguard against errors in constants, the statements listed in Table 2 were tested numerically, with exact rational arithmetic where the statement is exact; the scripts (Python) are distributed with the paper as ancillary files. In checks 3 and 4 the output of Algorithm 1, run with its true parameters, is compared with an independent exact decision of $V_d^{\infty} \geq r$, obtained by enumerating policies and testing $\rho_C \geq 1$ through the leading principal minors of $I - M_{CC}$ in rational arithmetic.

Checks 1 and 2 test the two new ingredients separately. Check 3 tests Sections 5–8 and Proposition 4.3, not the procedure $\mathcal{L}$, which is replaced by an oracle. In check 4 the procedure

	What is tested	Instances	Arithmetic	Outcome
1	The test of Proposition 5.5(c): monotonicity of $\lfloor G \rfloor$ and $\lfloor G(t+Nj\mathbf{1}) \rfloor = \lfloor G(t) \rfloor + (N-1)j$	one two-state game, 300 random integer points	exact	confirmed
2	Theorem 6.2: the ratio $\lvert \theta v_i - \lambda_i \rvert / (\theta R)$	(i) 200 random matrices of order 2 to 5 (150 reducible, 50 irreducible), $N = 50$ and $N = 400$; (ii) chains of up to six classes with equal or nearly equal Perron roots, coupling weights between 10^{-6} and 10^{6}, N up to 2^{40}	(i) floating point; (ii) 80 digits	largest ratio (i) 0.042 (0.026 on the irreducible matrices); (ii) 0.076
3	Algorithm 1, with $\mathcal{L}$ replaced by an oracle that returns arbitrary labels subject only to the two implications used in the proof of Proposition 4.3	54 games with two or three Despot states: ties at the threshold, reducible chains with equal Perron roots, weights between 10^{-6} and 10^{6}, and 40 random games	exact, except the fixed point x used to build the oracle (120 digits)	all outputs agree with the exact decision
4	Algorithm 1 with the procedure $\mathcal{L}$ itself, for $n = 1$	five games	exact	all outputs agree with the exact decision

Table 2: Numerical checks. In check 2 the constant R is evaluated with a value of Ω that is at most the right-hand side of (5), so the ratios are upper bounds for the ratios with any admissible Ω.

is the one of Appendix A, except that the vector t^* of its width base case is not obtained by the iteration (3.3) of [9]: for $n = 1$ it equals the clipping to the box of $\max\{t \in \mathbb{Z} : \lfloor G(t) \rfloor \geq t\}$, which is computed once, exactly. None of these checks tests Assumption A, except check 4 in the degenerate case $n = 1$. For $n \geq 2$ the procedure $\mathcal{L}$ was not executed at the parameters of Theorem 1.1: by Proposition 9.1 its recursion tree is too large for that even on tiny games.

C Formalisation of the correctness part

Scope. The correctness part of Theorem 1.1 and the results it rests on have been formalised in Lean 4 with Mathlib. The running-time part has not: the formalisation contains no Turing machine, no count of steps, no input encoding and no length L, hence no statement on bit sizes. Table 3 gives the coverage result by result. The Lean sources are distributed with the paper as ancillary files; the versions, the names of the files and declarations, and the list of axioms used are given in the accompanying README. In particular, Theorem 6.2 is the declaration `theorem_6_2`, and the theorem stated below is `main_correctness_rational`.

The formal theorem. Let an entropy game with finitely many Despot and People states, at least one of them a Despot state, have weights $m_e = b_e/c_e$, let $r = r_1/r_2$, with positive integers b_e, c_e, r_1, r_2, let q and $a_{\max}$ be as in Section 2, and let the parameters be as in Section 8, except that s and d_0 are any integers satisfying the stated inequalities, not necessarily the least ones. Then the loop of Proposition 4.3, for the map G of Section 8 built from the weights m_e/r, has a terminating run; and every terminating run makes at most $128(d_0+1)$ calls of $\mathcal{L}$ and ends with a vector A such that, for every $d \in D$, $A_d \geq -2^{13}N$ if and only if $V_d \geq r$. Here V is any

vector satisfying the saddle-point condition of Section 2 for the given game (there is at most one), strategies being deterministic and history-dependent as in that section.

Hypotheses not proved in Lean. The formal theorem has two hypotheses besides the definition of V, and neither is proved in Lean. The first is Assumption A(a), for the index set D only. It is stated for a transcription of the procedure of Appendix A as an inductive relation between an input, an output and a number of nodes; "terminates" means that a run exists. The second is the existence theorem [2, Theorem 10] in the form in which this paper uses it, for the game with weights m_e/r: there are a vector and a pair of policies satisfying the saddle-point condition against all deterministic history-dependent strategies. The node bound of Lemma 4.4 has Assumption A(b),(c) as hypotheses, stated for the same transcription (the sentence of Lemma 3.4 of [9] on the length of a pivot pass being stated for completed passes); the first sentence of Proposition 9.1 uses the transcription only. The transcription was compared item by item with Section 3.2 of [9] twice, each time by an AI system (see the Statement on the use of AI); neither comparison found a difference. The formal loop is an execution relation over the real-valued map G; the computation of $\lfloor G \rfloor$ (Lemma 5.6) is outside it.

Result	Formalised	Remark
Section 2, normalisation and payments	partly	scaling of the value; $m_e/r = a_e/q$; $R_d^k(\sigma,\tau) = ((M^{\sigma\tau})^k \mathbf{1})_d$; boundedness of $R_d^k(\sigma,\tau)^{1/k}$
[2, Theorem 10]; Assumption A	no	hypotheses; the sentence $V_d^\infty = \lim_k (V_d^k)^{1/k}$ of the former is neither used nor stated
Lemmas 4.1, 4.2; Proposition 4.3	yes	
Lemma 4.4	partly	the bound on the number of nodes and $B_0 \le 2 + 6\log_2 n$; not the two sentences on the boxes of the tree and on the evaluations of $\lfloor F \rfloor$
Lemmas 4.5, 5.6	no	
Lemmas 5.1, 5.2, 5.3; Proposition 5.5	yes	Proposition 5.5(d) is proved for every Ω with $e^{-\Omega} \le m_{pj}$ and $\sum_j m_{pj} \le e^{\Omega}$; the instance $\Omega = \ln(n \max(a_{\max}, q))$ is not stated
Perron–Frobenius theorem, as used in Section 6; Lemma 6.1; Theorem 6.2	yes	with the constant R as printed; heights of classes are not used, and the sentence that the classes of height 0 are nontrivial is not formalised
Remark 6.3	partly	$\lambda_1 = \lambda_2 = 0$, $v_2 = 0$ and the equation for X; not the asymptotic
Lemmas 7.1, 7.2	yes	with the constant H as printed
Section 8, parameters	partly	the inequalities $\|\tilde{f} - f\|_\infty \le 1/(64H)$ and $\theta R \le 1/(64H)$
Theorem 1.1, correctness	yes	in the form of the formal theorem above
Theorem 1.1, running time; the remarks following the theorem	no	
Proposition 9.1	partly	the first sentence and the inequality $\binom{n+B_0}{B_0} \ge (n/(7\log_2 n))^{5\log_2 n}$ for $n \ge 2^7$; not the sentence on the first call of the algorithm, nor the word "superpolynomial"

Table 3: Coverage of the formalisation.

Differences from the written proofs. The formal statements and all constants are those printed here, apart from generalisations. In three places the formal proof takes a different route, so that the written step is not what was checked. In the upper bounds of Lemma 6.1(d) and of Theorem 6.2 the induction is on the number of states accessible from i, not on the height of a

class, and in the former no vector u is built. In Lemma 4.4 the double sum is obtained through a recurrence, not by counting paths. Details, and smaller deviations of the same kind, are listed in the README.

Statement on the use of AI

This paper was produced with substantial use of generative AI, and the reader should weigh it accordingly. The author chose the problem area (mean-payoff and related games), decided to build on the OpenAI Math release and to treat its statements as hypotheses, set the procedure by which the work was done and checked, and made the decisions on scope. The specific question treated here (entropy games) was proposed by the AI system. The statements, the proofs, the computational checks and their scripts, the literature search, and the text of the paper were produced by instances of Anthropic's Claude (Claude Opus 5.5), run as autonomous agents under the author's direction. The checking was also done by AI: before posting, the paper was reviewed internally over several rounds by separate instances of the same model, its citations were compared with the cited sources by further Claude instances, most of them of a smaller Claude model, and the Lean formalisation described in Appendix C was written by such instances. The comparisons reported in that appendix, of the Lean statements with this paper and of the transcription of the procedure with [9], were likewise made by instances of the AI system and not by a person. The author also obtained comments on the paper from GPT-6 (OpenAI), which is again a review by an AI system. A later assessment of originality and impact by the same model led to the following changes: the sentence of the abstract stating which ingredients do not depend on the assumed theorem, the sentence of the Introduction disclaiming practical efficiency, and the names of the two Lean declarations in Appendix C. That assessment was not a verification of the proofs. The paper has not been refereed by a human. No AI system is an author; the author takes responsibility for the contents.

References

[1] E. Asarin, J. Cervelle, A. Degorre, C. Dima, F. Horn, V. Kozyakin. Entropy games and matrix multiplication games. In *STACS 2016*; arXiv:1506.04885v3.

[2] M. Akian, S. Gaubert, J. Grand-Clément, J. Guillaud. The operator approach to entropy games. *Theory of Computing Systems* 63(5):1089–1130, 2019; arXiv:1904.05151v1.

[3] X. Allamigeon, S. Gaubert, R. D. Katz, M. Skomra. Universal complexity bounds based on value iteration for stochastic mean payoff games and entropy games. arXiv:2206.09044v2, 2024; journal version: *Information and Computation* 302:105236, 2025. Statements are cited by their numbering in the arXiv version.

[4] M. Akian, S. Gaubert, U. Naepels, B. Terver. Solving irreducible stochastic mean-payoff games and entropy games by relative Krasnoselskii–Mann iteration. arXiv:2305.02458v1, 2023.

[5] C. Baier, K. Chatterjee, T. Meggendorfer, J. Piribauer. Entropic risk for turn-based stochastic games. arXiv:2307.06611v1, 2023 (extended version of an MFCS 2023 paper).

[6] W. Czerwiński, L. Daviaud, N. Fijalkow, M. Jurdziński, R. Lazić, P. Parys. Universal trees grow inside separating automata: quasi-polynomial lower bounds for parity games. In *SODA 2019*; arXiv:1807.10546v2.

[7] K. Etessami, A. Stewart, M. Yannakakis. A note on the complexity of comparing succinctly represented integers, with an application to maximum probability parsing. arXiv:1304.5429v3, 2014.

[8] N. Fijalkow, P. Gawrychowski, P. Ohlmann. The complexity of mean payoff games using universal graphs. arXiv:1812.07072v2, 2019.

[9] OpenAI. Turn-based stochastic mean-payoff games in deterministic quasipolynomial time. Manuscript dated October 5, 2026, in [11].

[10] OpenAI. Deterministic quasipolynomial-time mean-payoff games. Manuscript dated September 25, 2026, in [11].

[11] OpenAI. OpenAI Math release. https://github.com/openai/math, commit `fd4aeeb`, October 2026.